Engineering Mechanics: ME101

Lecture 8 : Statics

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Theorems of Pappus-Guldinus

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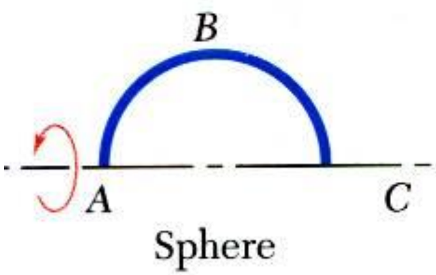


Figure (2)

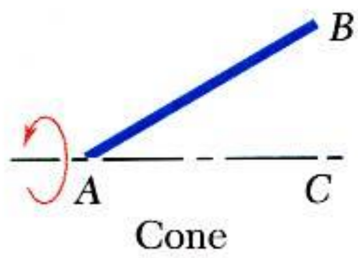
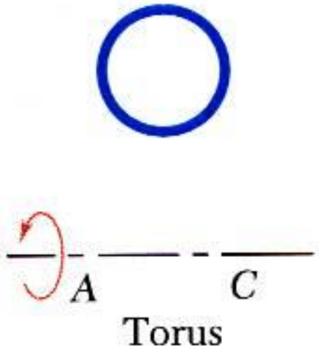
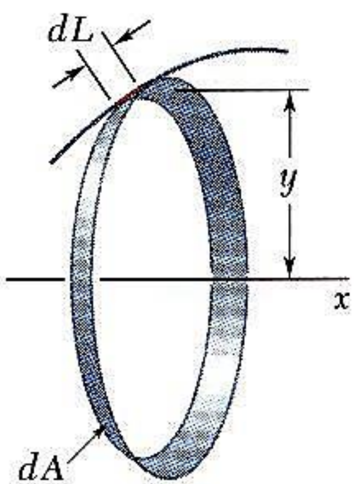


Figure (3)



- **Surface of revolution** is generated by rotating a plane curve about a fixed axis.



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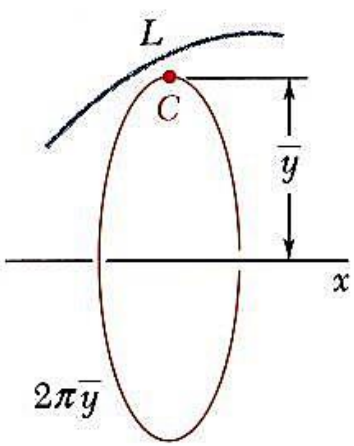


Figure (5)

❖ Area of a surface of revolution is equal to the length of the generating curve times the distance traveled by the centroid through the rotation.

$$A = 2\pi \bar{y} L \qquad \textbf{(i)}$$

Theorems of Pappus-Guldinus

Figure (1)

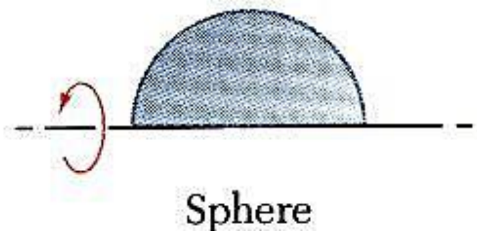
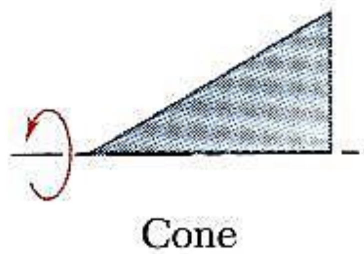


Figure (2)



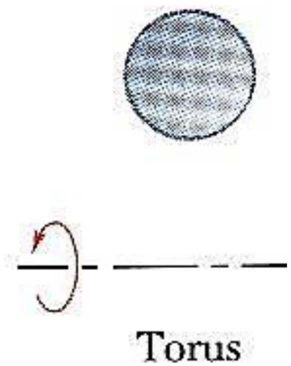


Figure (3)

- **Body of revolution** is generated by rotating a plane area about a fixed axis.

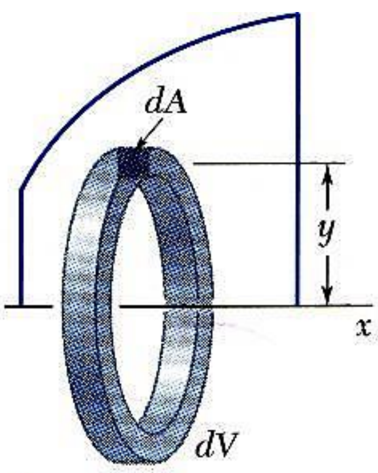


Figure (4)

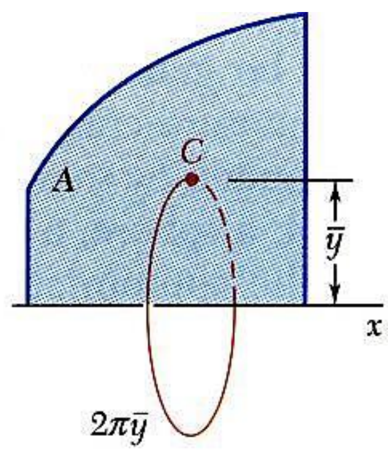


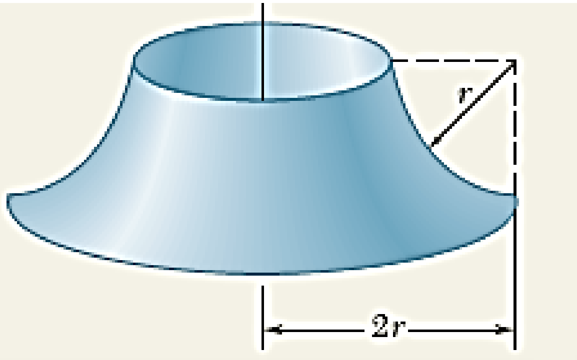
Figure (5)

❑ Pappus-Guldinus 2nd Theorem:

❖ Volume of a body of revolution is equal to the generating area times the distance traveled by the centroid through the rotation.

$$V = 2\pi\bar{y}A \quad \textbf{(i)}$$

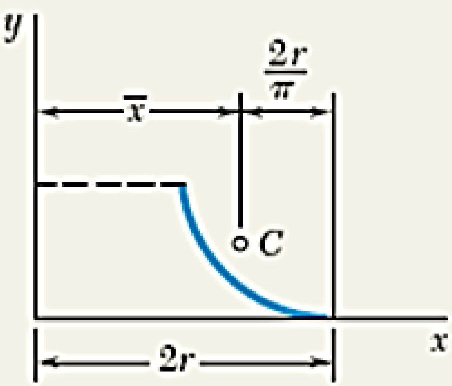
Sample Problem



Determine the area of the surface of revolution shown, which is obtained by rotating a quarter-circular arc about a vertical axis.

Figure (1)

SOLUTION



According to Theorem I of Pappus-Guldinus, the area generated is equal to the product of the length of the arc and the distance traveled by its centroid.

$$\bar{x} = 2r - \frac{2r}{\pi} = 2r\left(1 - \frac{1}{\pi}\right) \quad \text{(i)}$$

$$A = 2\pi\bar{x}L = 2\pi\left[2r\left(1 - \frac{1}{\pi}\right)\right]\left(\frac{\pi r}{2}\right) \quad \text{(ii)}$$

Figure (2)

$$\boxed{A = 2\pi r^2(\pi - 1)} \quad \text{(iii)}$$

Sample Problem

Using the theorems of Pappus-Guldinus, determine (*a*) the centroid of a semicircular area, (*b*) the centroid of a semicircular arc. We recall that the volume and the surface area of a sphere are $\frac{4}{3}\pi r^3$ and $4\pi r^2$, respectively.

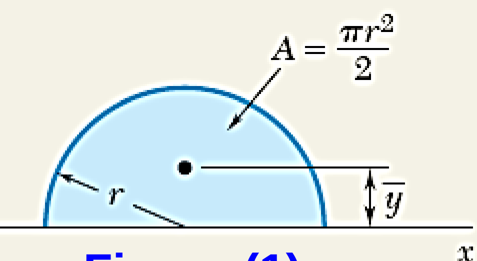


Figure (1)

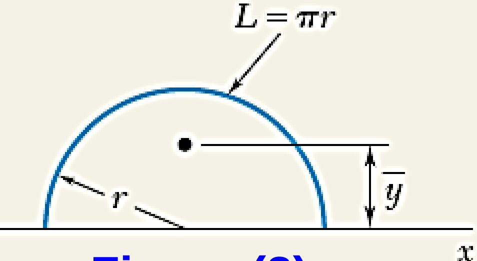


Figure (2)

SOLUTION

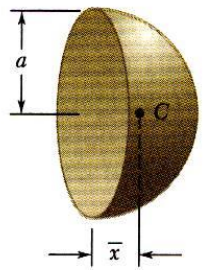
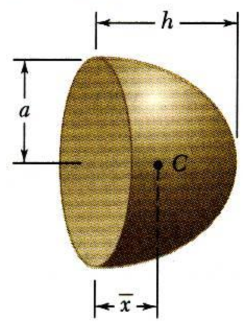
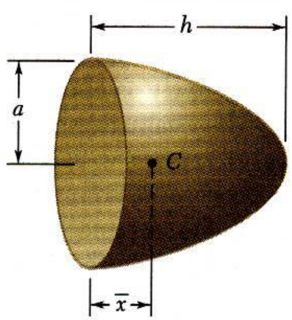
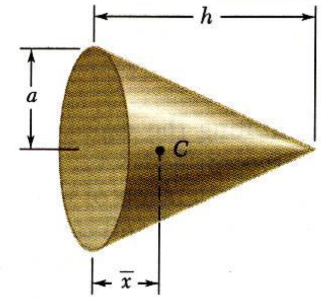
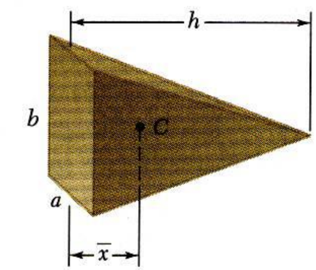
The volume of a sphere is equal to the product of the area of a semicircle and the distance traveled by the centroid of the semicircle in one revolution about the x axis.

$$V = 2\pi\bar{y}A \quad \textbf{(i)} \quad \tfrac{4}{3}\pi r^3 = 2\pi\bar{y}\left(\tfrac{1}{2}\pi r^2\right) \quad \textbf{(ii)} \quad \bar{y} = \frac{4r}{3\pi} \quad ◀$$

Likewise, the area of a sphere is equal to the product of the length of the generating semicircle and the distance traveled by its centroid in one revolution.

$$A = 2\pi\bar{y}L \quad \textbf{(iii)} \quad 4\pi r^2 = 2\pi\bar{y}(\pi r) \quad \textbf{(iv)} \quad \bar{y} = \frac{2r}{\pi} \quad ◀$$

Centroids of Common 3D Shapes

Shape		$\bar{x}$	Volume
Hemisphere		$\frac{3a}{8}$	$\frac{2}{3}\pi a^3$
Semiellipsoid of revolution		$\frac{3h}{8}$	$\frac{2}{3}\pi a^2 h$
Paraboloid of revolution		$\frac{h}{3}$	$\frac{1}{2}\pi a^2 h$
Cone		$\frac{h}{4}$	$\frac{1}{3}\pi a^2 h$
Pyramid		$\frac{h}{4}$	$\frac{1}{3}abh$

Distributed loads on Beams

- Determination of Resultant Force (*R*) on beam is important

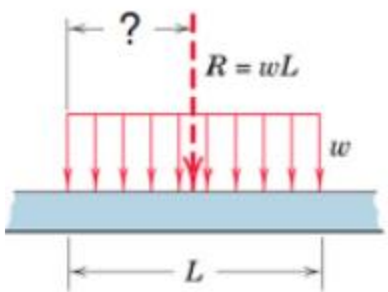


Figure (1)

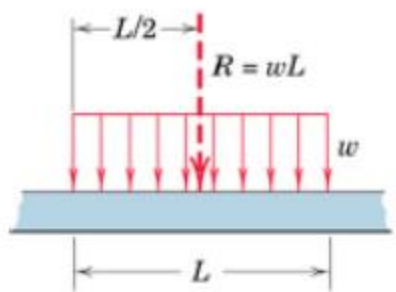


Figure (2)

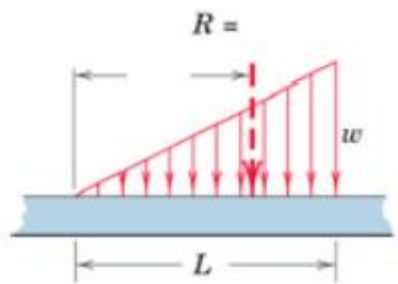


Figure (3)

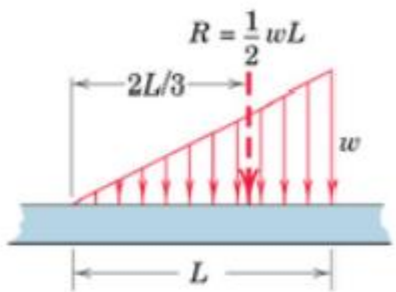


Figure (4)

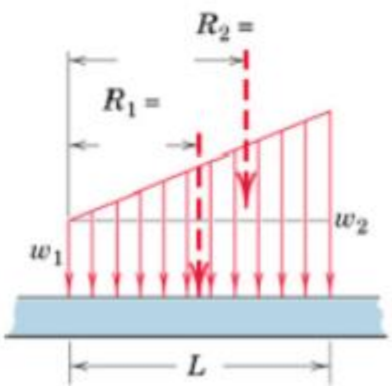


Figure (5)

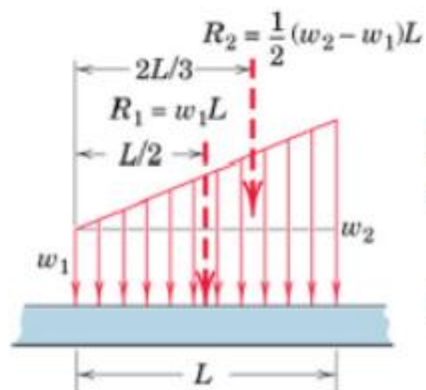


Figure (6)

R = area formed by *w* and length *L* over which the load is distributed

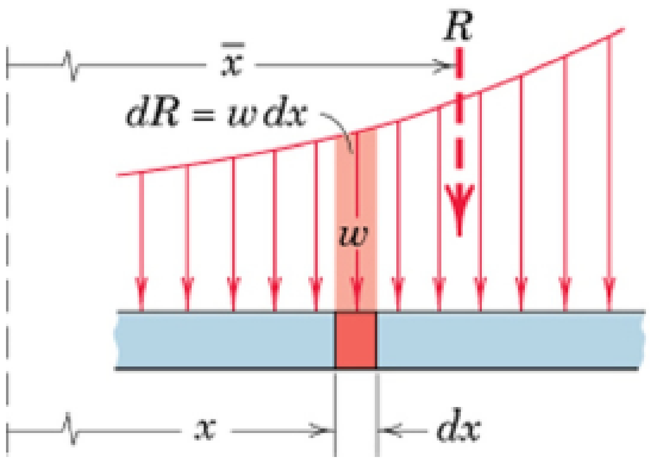
R passes through centroid of this area

- ❑ **Note:** where '*w*' is the load per unit length
- ❑ To find the overall Centroid of distributed load we will use Moment of Sum = Sum of Moment.

If the variable load is acting along a Curved Surface

Distributed Loads on beams

- General Load Distribution



Differential increment of force is

$dR = w\,dx$ **(i)**

Total load R is sum of all the differential forces

$R = \int w\,dx$ acting at centroid of the area under consideration

$$\bar{x} = \frac{\int xw\,dx}{R}$$ **(ii)**

Once R is known reactions can be found out from Statics

Distributed Loads on Beams

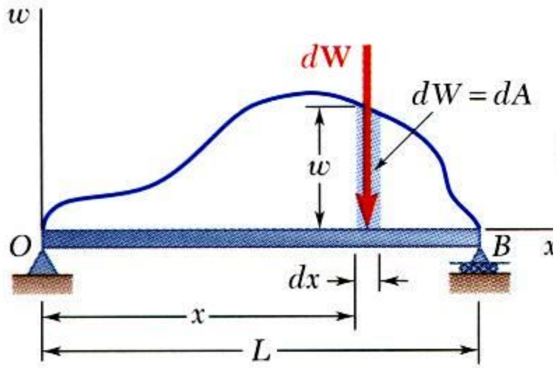


Figure (1)

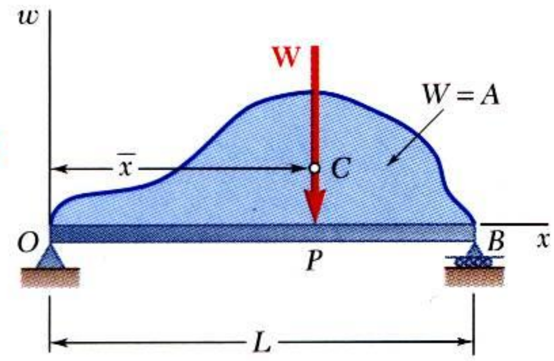


Figure (2)

$$W = \int_0^L w\,dx = \int dA = A \qquad \textbf{(i)}$$

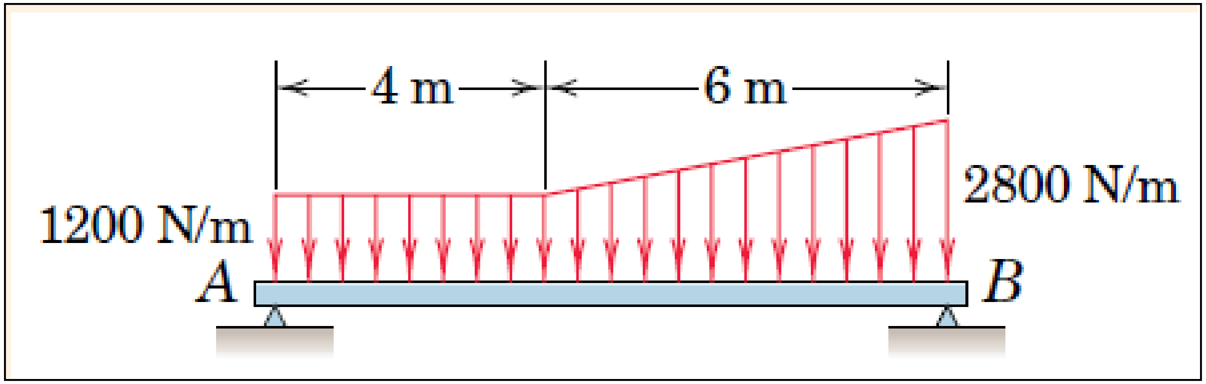
1. A distributed load is represented by plotting the load per unit length, *w* (N/m) . The total load is equal to the area under the load curve.

$$(OP)W = \int x\,dW \qquad \textbf{(ii)}$$

$$(OP)A = \int_0^L x\,dA = \bar{x}A \qquad \textbf{(iii)}$$

2. A distributed load can be replace by a concentrated load with a magnitude equal to the area under the load curve and a line of action passing through the area centroid.

(Problem-1) Determine the equivalent concentrated load(s) and external reactions for the simply supported beam which is subjected to the distributed load shown in **Fig(a)**?



Fig(a)

Solution of Problem1 *(contd.)*

❖ Converting the 2800 N/m and 1200 N/m variable loads to Point loads we have,

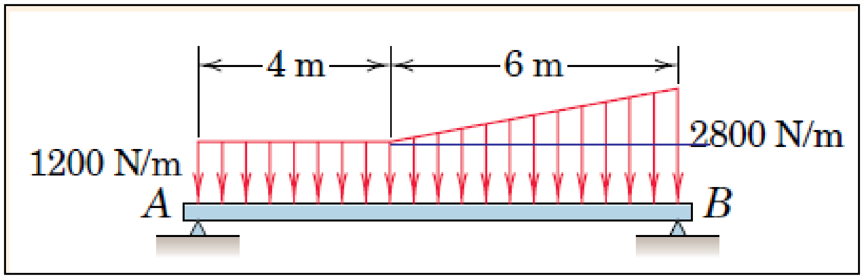


Figure (1)

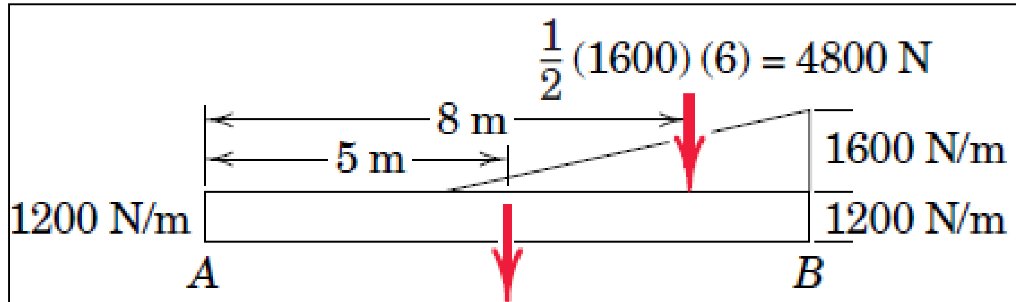


Figure (2)

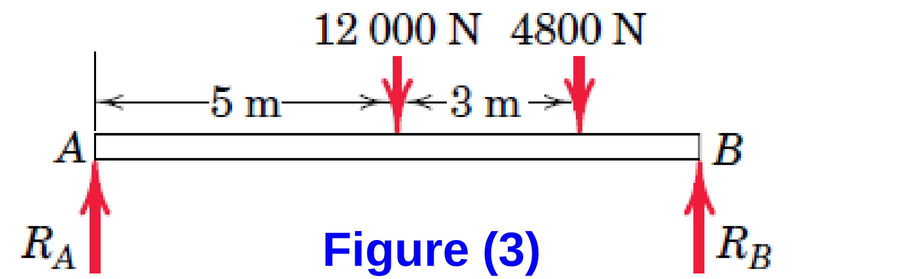


Figure (3)

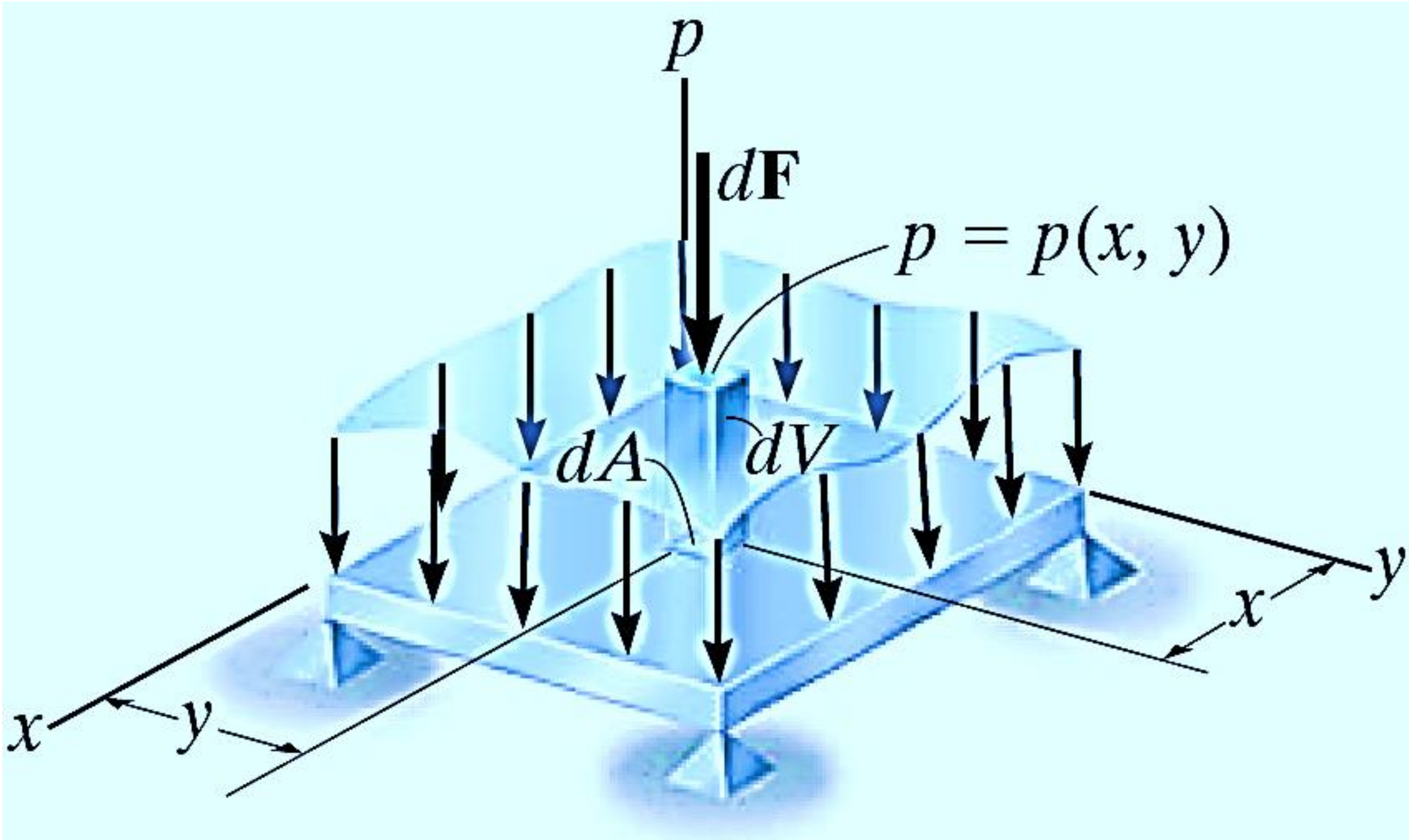
$$[\Sigma M_A = 0] \qquad 12\,000(5) + 4800(8) - R_B(10) = 0 \qquad \textbf{(i)}$$

$$R_B = 9840 \text{ N or } 9.84 \text{ kN}$$

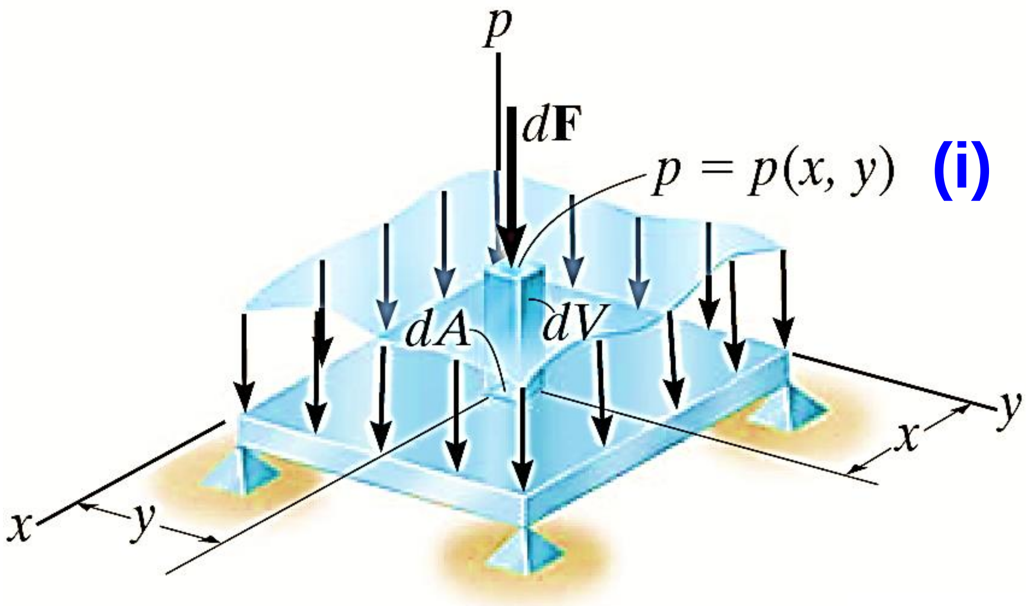
$$[\Sigma M_B = 0] \qquad R_A(10) - 12\,000(5) - 4800(2) = 0 \qquad \textbf{(ii)}$$

$$R_A = 6960 \text{ N or } 6.96 \text{ kN}$$

Distributed forces: Pressure load on a flat plate



Distributed forces: Pressure load on a flat plate



(i)

Figure (1)

$$F_R = \int_A p(x, y)\, dA$$

(ii)

$$\bar{x} = \frac{\int_A x p(x, y)\, dA}{\int_A p(x, y)\, dA} \qquad \bar{y} = \frac{\int_A y p(x, y)\, dA}{\int_A p(x, y)\, dA}$$

(iii)

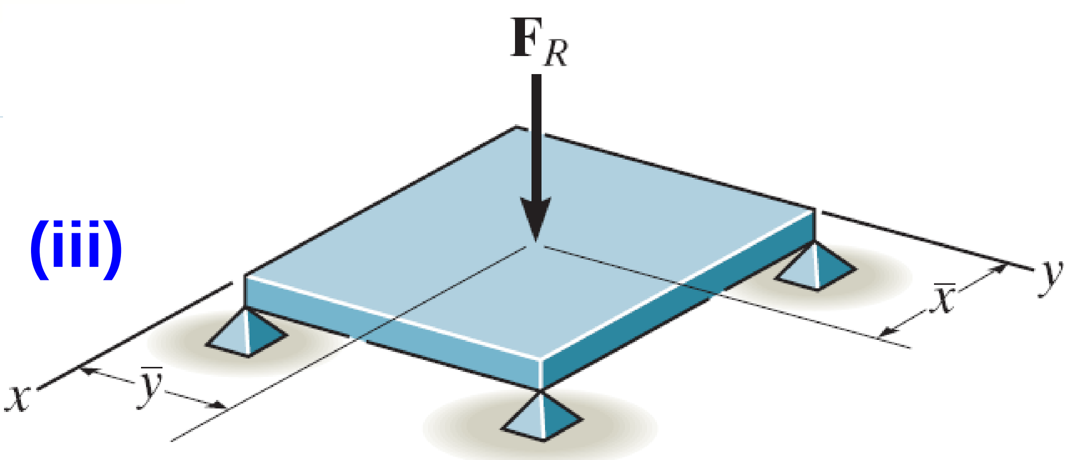
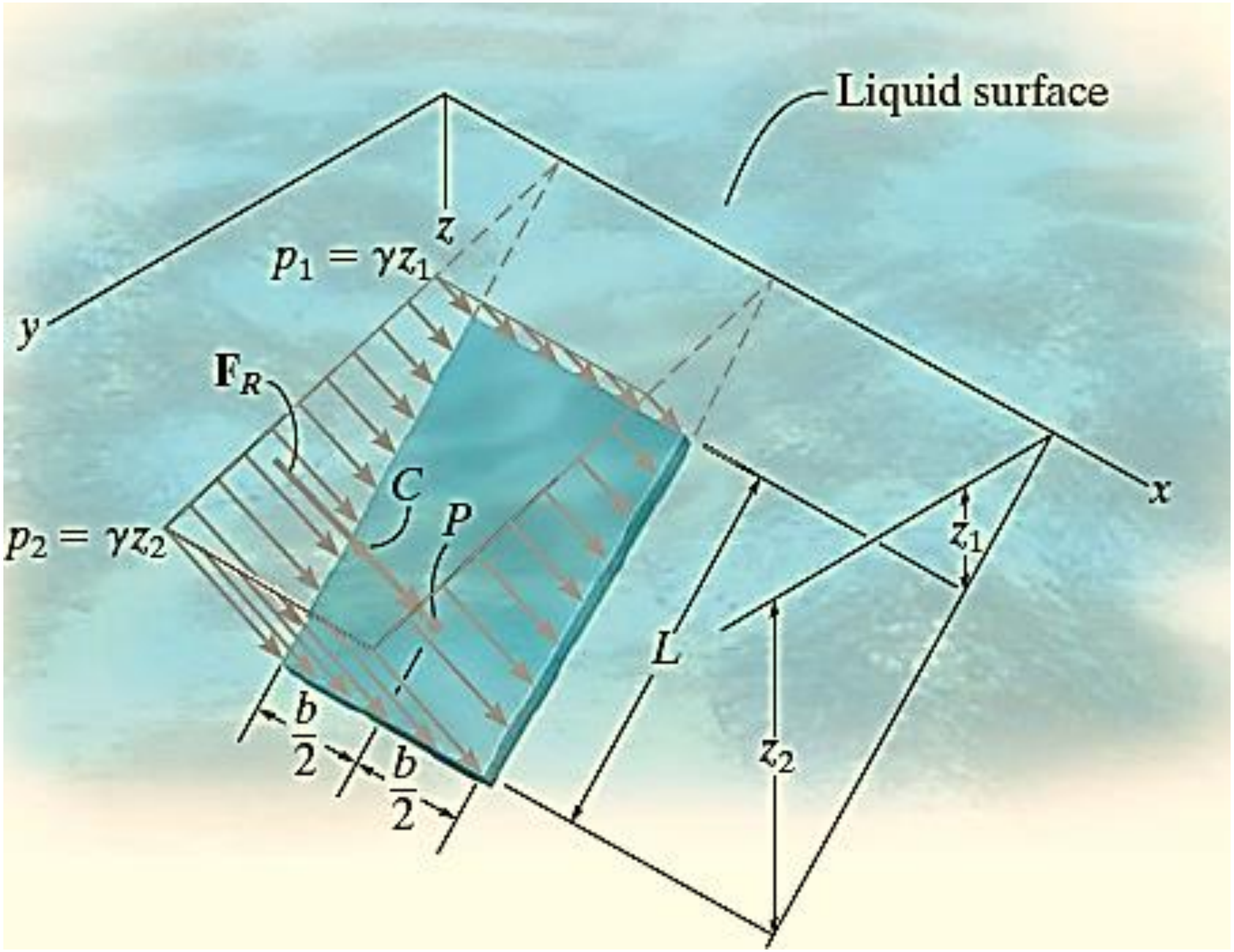


Figure (2)

Distributed forces: Hydrostatic forces



Hydrostatics of submerged bodies

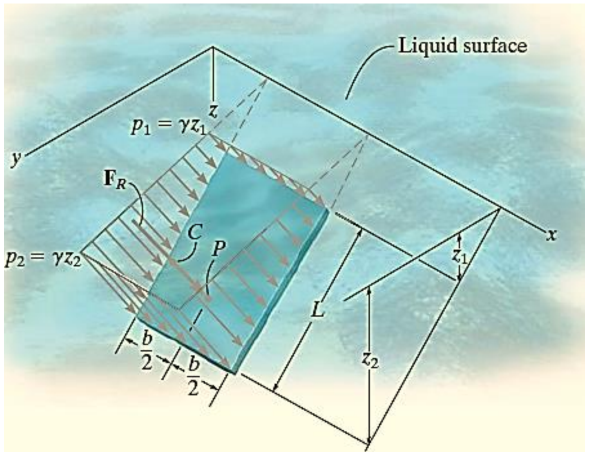


Figure (1)

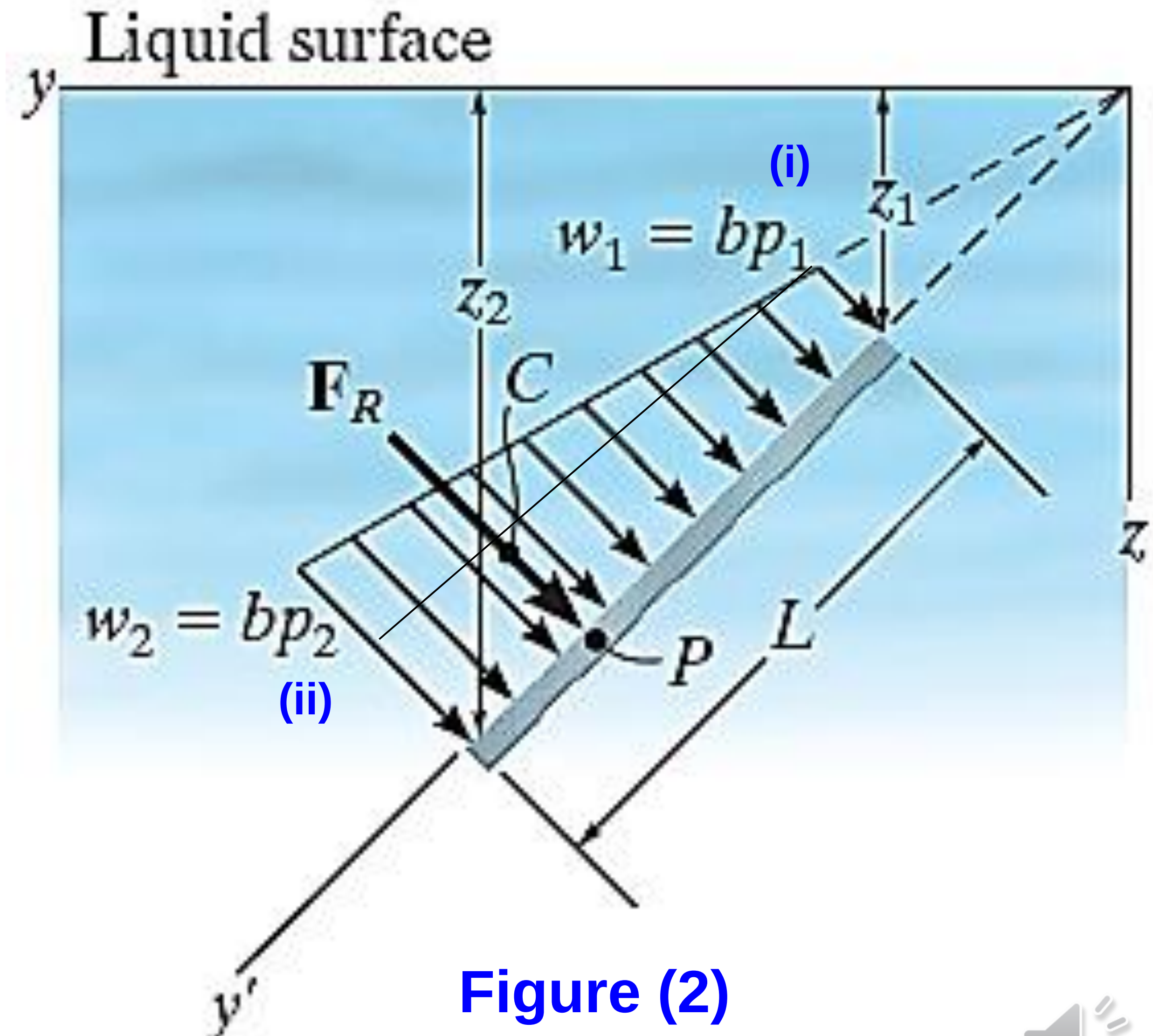


Figure (2)

Hydrostatics of submerged bodies

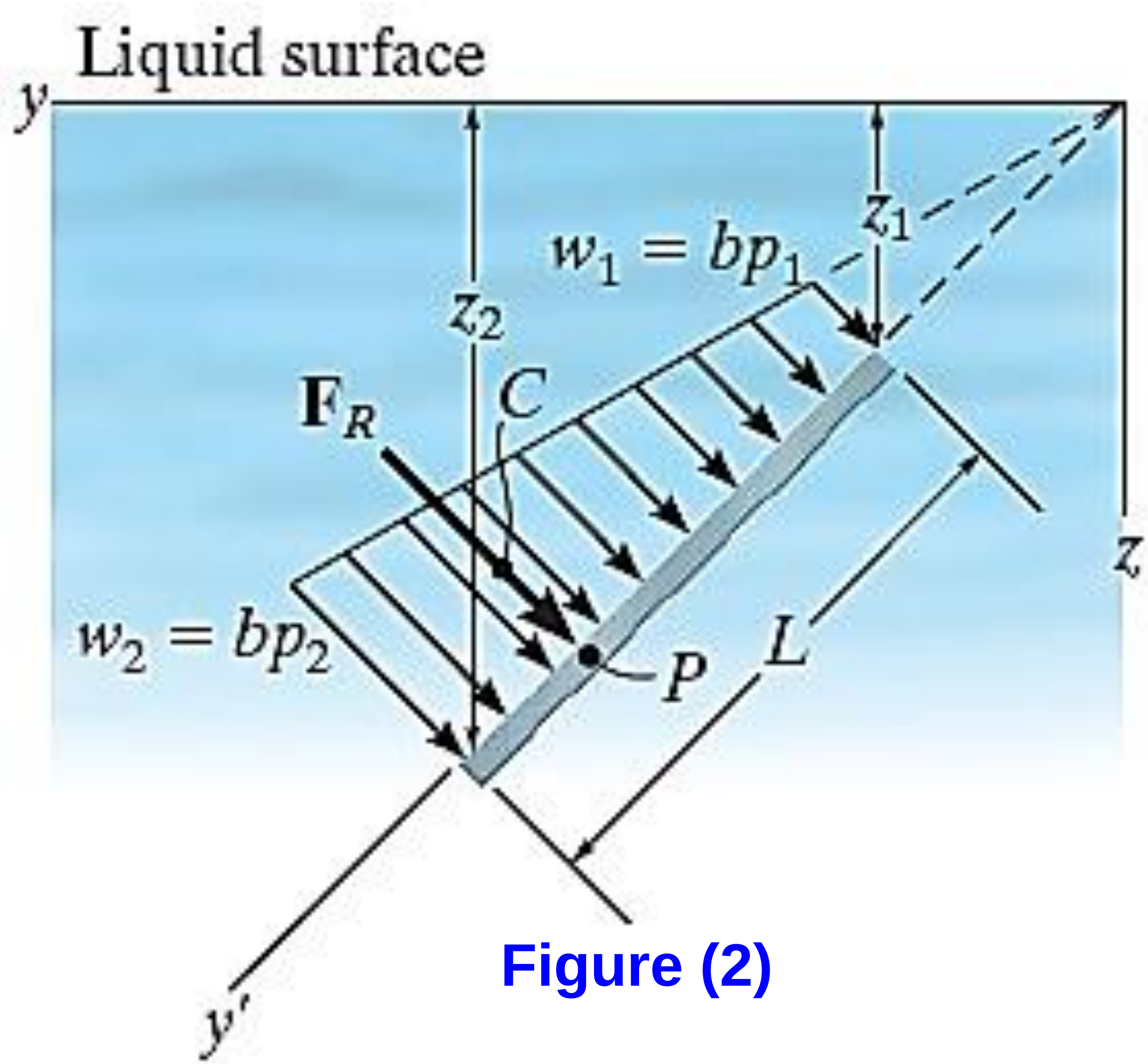
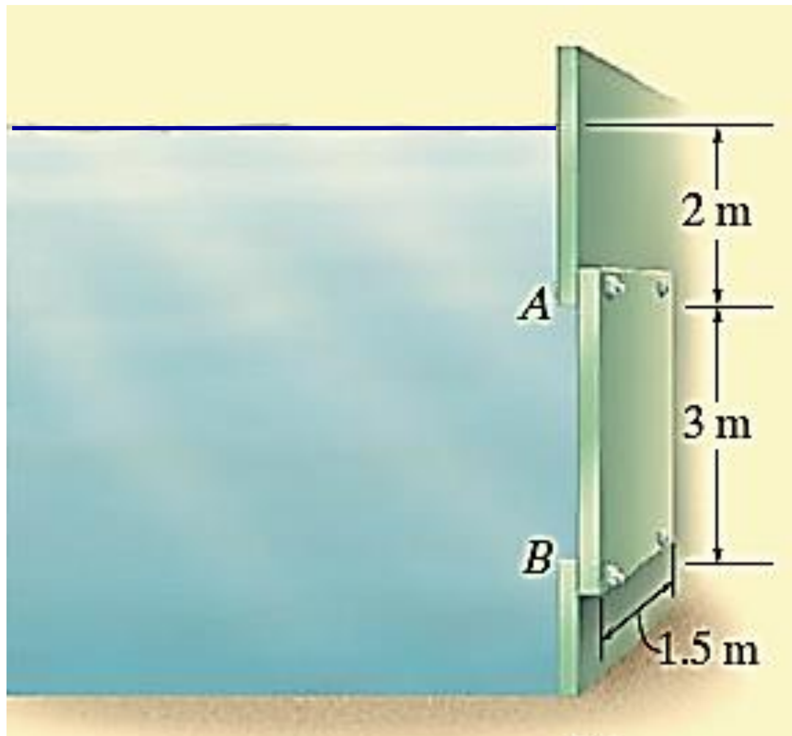


Figure (2)

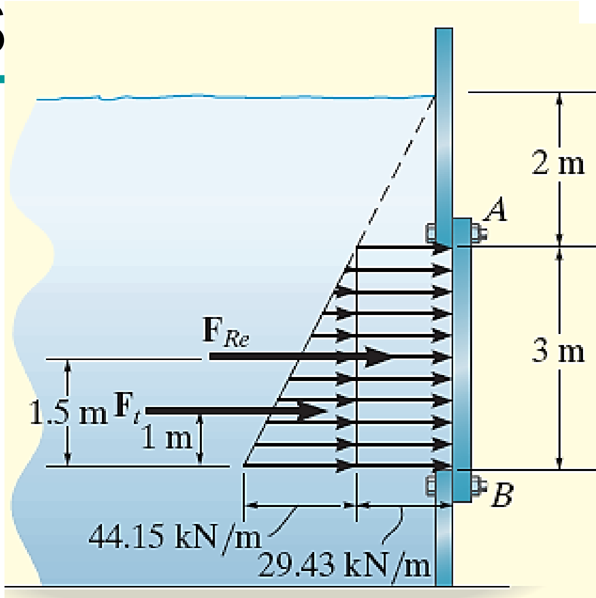
Sample problem: Hydrostatics

Determine the magnitude and location of the resultant hydrostatic force acting on the submerged rectangular plate AB shown in Fig. a. The plate has a width of 1.5 m; $\rho_w = 1000\ \mathrm{kg/m^3}$.



(a)

Sample problem: Hydrostatics



The water pressures at depths A and B are

$$p_A = \rho_w g z_A = (1000\ \mathrm{kg/m^3})(9.81\ \mathrm{m/s^2})(2\ \mathrm{m}) = 19.62\ \mathrm{kPa} \quad \textbf{(i)}$$

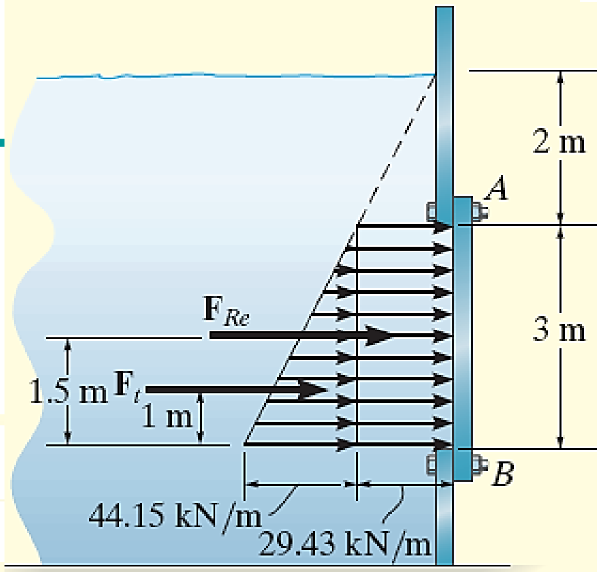
$$p_B = \rho_w g z_B = (1000\ \mathrm{kg/m^3})(9.81\ \mathrm{m/s^2})(5\ \mathrm{m}) = 49.05\ \mathrm{kPa} \quad \textbf{(ii)}$$

Since the plate has a constant width, the pressure loading can be viewed in two dimensions as shown in Fig. The intensities of the load at A and B are

$$w_A = bp_A = (1.5\ \mathrm{m})(19.62\ \mathrm{kPa}) = 29.43\ \mathrm{kN/m} \quad \textbf{(iii)}$$

$$w_B = bp_B = (1.5\ \mathrm{m})(49.05\ \mathrm{kPa}) = 73.58\ \mathrm{kN/m} \quad \textbf{(iv)}$$

Sample problem: Hydrostatics



Consider two components of $\mathbf{F}_R$, defined by the triangle and rectangle shown in Fig. Each force acts through its associated centroid and has a magnitude of

$$F_{Re} = (29.43 \text{ kN/m})(3 \text{ m}) = 88.3 \text{ kN} \quad \textbf{(v)}$$

$$F_t = \tfrac{1}{2}(44.15 \text{ kN/m})(3 \text{ m}) = 66.2 \text{ kN} \quad \textbf{(vi)}$$

Hence,

$$F_R = F_{Re} + F_t = 88.3 + 66.2 = 154.5 \text{ kN} \qquad \textit{Ans.}$$

The location of $\mathbf{F}_R$ is determined by summing moments about B,

$$\circlearrowleft +(M_R)_B = \Sigma M_B;\ (154.5)h = 88.3(1.5) + 66.2(1) \quad \textbf{(vii)}$$

$$h = 1.29 \text{ m} \qquad \textit{Ans.}$$

END